
Calibration Is Not Stability: A DFT-Anchored Compact State Map for Layered Interatomic Potentials

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Abstract

1 Layered interatomic potentials introduce electronic-state-dependent forces by mix-
2 ing fixed cold and hot models, yet their state controller is usually not identified
3 from intermediate-temperature electronic structure. We introduce CALM, a caloric,
4 accuracy-calibrated, least-motion controller: three electronic- temperature anchors
5 are joined by a compact C^2 minimum-jerk map, its middle anchor minimizes a
6 dimensionless energy-force loss, an equilibrium quantile sets an exact cold guard,
7 and its remaining dynamical scale is fixed by matching integrated excitation to a de-
8 clared sigmoid. On 2444 configurations withheld from both endpoint fits, CALM
9 reduces energy RMSE from 29.02 to 12.16 meV/atom and force RMSE from
10 0.0924 to 0.0880 eV/Å. Ninety calibration trajectories match sigmoid exposure
11 within 5.9%, while only 0.0100% of 2073600 equilibrium atom-frames activate.
12 The harder tests are negative. In a paired 1,024-atom recoil stress experiment,
13 CALM becomes unevaluable at 15 keV, one energy before the exposure-matched
14 sigmoid, so compact tails do not extend the registered stable range. Of 1450
15 selected continuations, 1432 reach 30 ps; all 18 failures reproduce under exact
16 replay. CALM completes 95/100 trajectories, versus 114/ 125 for the matched
17 sigmoid. Among completed pairs, CALM yields fewer surviving Frenkel pairs than
18 ground ACE at only 1/4 energies, although its peak disordered volume is smaller at
19 3/4. CALM therefore provides a traceable interpolation calibration but fails both
20 dynamical hypotheses, showing that exposure equality and compact support are
21 insufficient controller criteria for this nonconservative force blend.

22 1 Introduction

23 Classical molecular dynamics (MD) obtains its reach by integrating nuclear motion on a fixed Born-
24 Oppenheimer potential-energy surface. That reduction is often excellent near equilibrium, but it
25 is least secure during an irradiation cascade: energetic atoms transfer energy to both nuclei and
26 electrons, the two subsystems need not share a temperature, and electronic occupation can alter the
27 forces while the collision cascade is still evolving [Duffy and Rutherford, 2007, Zarkadoula et al.,
28 2015, Zhang and Weber, 2020]. This is a particularly consequential approximation for tungsten,
29 where primary-damage predictions support the design of plasma-facing components and where the
30 hot, disordered cascade core spans conditions far from those represented by a ground-state fit.

31 Two complementary lines of work address this missing physics. Two-temperature MD evolves
32 a coarse-grained electronic energy field and exchanges energy with the atoms through electronic
33 stopping, electron-phonon coupling, diffusion, and stochastic forces [Duffy and Rutherford, 2007,
34 Zarkadoula et al., 2015, Sand et al., 2025]. Electronic-temperature-dependent potentials instead
35 change the interatomic force law itself, using finite-temperature electronic structure to represent the
36 redistribution of electronic occupations [Mermin, 1965, Khakshouri et al., 2008, Ackland, 2012,
37 Murphy et al., 2016]. The two mechanisms are not substitutes: an energy exchange model need not

change the potential surface, while a state-dependent surface need not conserve or transport electronic energy.

Machine-learned interatomic potentials make the second route practical at cascade length scales. In particular, the atomic cluster expansion (ACE) provides a systematically improvable local representation with efficient force evaluation [Drautz, 2019], and FitSNAP connects such representations to electronic-structure data and LAMMPS [Rohskopf et al., 2023, Thompson et al., 2022]. Wood, Koknat, and Thompson introduced a layered construction in which two fixed ACE models—one trained near the electronic ground state and one at elevated electronic temperature—are mixed per atom through LAMMPS *hybrid/scaled* [Wood et al., 2026]. Their radiation-damage realization used a smooth sigmoid of center-of-mass-corrected local ionic temperature. It established a scalable way to expose an MD trajectory to more than one electronic surface, but left the sigmoid center and width as physical parameters rather than quantities identified from the intermediate-temperature electronic-structure data.

The controller is not a cosmetic part of a layered model. It must decide which mixture represents an intermediate electronic state, infer that state from an ionic observable, reject equilibrium thermal fluctuations, and control the time spent on a force field that is generally nonconservative under atom-wise mixing. A sigmoid assigns all four jobs to its center and width. Its infinite tails also retain nonzero mixed-force exposure after a thermal spike has cooled, even when the integrated excitation is small.

We introduce the *caloric, accuracy-calibrated, least-motion* (CALM) map, which separates these decisions into auditable calibrations. First, a single intermediate blend fraction is selected by a dimensionless joint energy-force criterion on electronic-temperature data withheld from both endpoint fits. Second, the three electronic-temperature anchors are joined by the unique minimum-integrated-squared-jerk quintic, whose compact extension is twice continuously differentiable and reaches exact cold and hot plateaus. Third, a Sommerfeld-motivated caloric transform maps local ionic kinetic energy to an electronic-temperature proxy. A high quantile of equilibrium fluctuations sets a cold guard, and the only remaining scale is fixed by matching integrated high-temperature residency to a declared sigmoid comparator without consulting stability or damage outcomes. The construction therefore distinguishes *accuracy calibration* from *exposure matching*; neither the two ACE models nor their force-layering algebra is changed.

The evaluation gives a deliberately split answer. CALM improves both held-out energy and force errors and nearly eliminates equilibrium activation, but its matched-exposure recoil series suffers a reproducible missing trajectory at 15 keV; the comparator remains evaluable through 15 keV and becomes incomplete at 20 keV. Compact support therefore solves an identification and false-tail problem without solving the layered force law’s dynamical instability. A checkpoint-exact 30 ps continuation tests whether the reported reduction of damage relative to ground ACE survives over CALM’s four selected recoil energies. All 18 failed continuations reproduce under exact replay, and the surviving complete-case contrasts preserve the spike-volume direction but not the Frenkel-pair direction at a majority of energies. Because compute measurements forced that campaign into a 1,024-atom cell, we interpret both negative outcomes as a paired controller stress test and retain the source study’s large-cell curves as a separate, explicitly digitized baseline.

2 Relation to prior work

Finite-temperature force surfaces. Mermin’s variational formulation supplies the finite-temperature extension of density-functional theory (DFT) [Mermin, 1965]. In a metal, a Sommerfeld expansion gives a leading electronic free-energy correction proportional to T_e^2 when the density of states is regular near the Fermi level. Ackland embedded that dependence in an empirical potential through an explicit entropy term [Ackland, 2012]. Electron-temperature-dependent tungsten potentials were subsequently fitted to finite-temperature DFT and used in laser-driven two-temperature simulations [Khakshouri et al., 2008, Murphy et al., 2016]. Those approaches parameterize a continuous temperature-dependent model. CALM addresses a different setting: two already-trained limiting models are held fixed, and the interpolation rule is identified from an independently supplied intermediate-temperature archive.

Electronic energy exchange during cascades. The two-temperature cascade model of Duffy and Rutherford couples MD to an electronic heat equation, returning energy through a local-temperature thermostat [Duffy and Rutherford, 2007]. Later tungsten calculations showed that the combination of electronic stopping and electron-phonon coupling can change both transient and residual damage [Zarkadoulas et al., 2015]; recent density-dependent coupling work reduces reliance on separate, thresholded dissipation terms [Sand et al., 2025]. Reviews emphasize that electronic and nuclear energy deposition can be competitive, additive, or synergistic rather than universally reducible to one friction coefficient [Zhang and Weber, 2020]. CALM does not evolve an electronic heat equation and is not offered as a replacement for stopping or electron-phonon transport. Its narrower purpose is to make the state-selection layer of a two-surface force model data-calibrated and protected by an empirical equilibrium guard.

Machine-learned and adaptively mixed potentials. ACE organizes local invariant basis functions by body order and polynomial degree [Drautz, 2019]; related tungsten machine-learned potentials have demonstrated that DFT-informed local models can support large radiation-damage simulations [Wood et al., 2019]. Layered Sommerfeld ACE extends this idea by retaining two finite-electronic-temperature fits and applying a dynamic, per-atom scale factor [Wood et al., 2026]. Adaptive-precision potentials provide an alternative route in which switching and its derivative terms are built into a conservative energy model, at the price of model-specific implementation [Immel et al., 2025]. CALM deliberately retains the general force-level LAMMPS mechanism of the layered construction. Consequently, its compact tails reduce a hypothesized source of unnecessary mixed-force residency, but do not make the dynamics conservative; that limitation is made explicit in Section 3.

3 The CALM map

3.1 Fixed force layering and design scope

Let $\mathcal{U}_g$ and $\mathcal{U}_h$ denote the supplied ground- and high-electronic-temperature ACE models, trained at

$$T_L = 0.001 \text{ eV}, \quad T_H = 0.5 \text{ eV}, \quad (1)$$

respectively. Their parameters, descriptors, and training data are not modified. For atom i , the layered force is

$$\mathbf{F}_i^{\text{lay}} = [1 - \lambda_i] \mathbf{F}_{g,i} + \lambda_i \mathbf{F}_{h,i} + \mathbf{F}_i^{\text{ZBL}}, \quad (2)$$

where the common short-range Ziegler-Biersack-Littmark contribution is unscaled [Ziegler et al., 2010]. Equation (2) and both ACE files are held fixed; only the map producing $\lambda_i \in [0, 1]$ is designed.

CALM factors that map as

$$\lambda_i = q(\hat{T}_{e,i}), \quad (3)$$

where q is calibrated against finite-electronic-temperature DFT and $\hat{T}_{e,i}$ is a caloric proxy inferred from the same local ionic temperature used by the sigmoid comparator. This factorization prevents an ionic switching threshold from being interpreted as though it were itself an electronic smearing temperature.

3.2 A compact C^2 electronic-state interpolant

For $x \in \mathbb{R}$, define

$$S(x) = \begin{cases} 0, & x \leq 0, \\ 6x^5 - 15x^4 + 10x^3, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases} \quad (4)$$

The interior polynomial is the standard minimum-jerk construction [Flash and Hogan, 1985], used here as an endpoint-flat interpolant rather than claimed as a new polynomial. Its role in a DFT-calibrated, guarded, exposure-matched composition is the contribution. Minimize $\mathcal{J}[p] = \int_0^1 [p'''(x)]^2 dx$ subject to

$$p(0) = p'(0) = p''(0) = 0, \quad p(1) = 1, \quad p'(1) = p''(1) = 0. \quad (5)$$

The Euler-Lagrange equation is $p^{(6)} = 0$, hence the minimizer is a quintic. The six endpoint conditions uniquely give $p(x) = 10x^3 - 15x^4 + 6x^5$. Since its first two derivatives vanish at both

endpoints, extending it by constants gives the compact C^2 function in Eq. (4). “Least motion” refers specifically to this minimum parametric jerk; it is not an assertion that an MD trajectory minimizes mechanical jerk.

Let $T_M = 0.1$ eV be the independently supplied intermediate electronic temperature and let λ_M be selected by the accuracy criterion below. The electronic-state map is

$$q(T_e) = \begin{cases} \lambda_M S\left(\frac{T_e - T_L}{T_M - T_L}\right), & T_e \leq T_M, \\ \lambda_M + (1 - \lambda_M) S\left(\frac{T_e - T_M}{T_H - T_M}\right), & T_e > T_M. \end{cases} \quad (6)$$

Thus $q(T_L) = 0$, $q(T_M) = \lambda_M$, and $q(T_H) = 1$ exactly. Both branches have zero first and second derivative at T_M , so they join in C^2 ; the constant tails preserve that regularity at T_L and T_H . Moreover, $S'(x) = 30x^2(1-x)^2 \geq 0$ on $[0, 1]$, so q is monotone for every $\lambda_M \in [0, 1]$.

3.3 Energy-force calibration of the middle anchor

For static validation, a configuration uses one global blend fraction λ . This special case has the well-defined interpolated predictions

$$E_{\lambda,k} = (1 - \lambda)\tilde{E}_{g,k} + \lambda\tilde{E}_{h,k}, \quad (7)$$

$$\mathbf{F}_{\lambda,k} = (1 - \lambda)\mathbf{F}_{g,k} + \lambda\mathbf{F}_{h,k}. \quad (8)$$

Here each endpoint prediction contains the same ZBL term; its affine blend therefore includes that term exactly once, consistent with Eq. (2). Every converged configuration k in the T_M archive is evaluated once with each immutable endpoint ACE model; subsequent values of λ are formed algebraically from those predictions. Files below any `NotConverged` path are excluded before parsing, and no structure is relaxed.

The distributed ACE files set their isolated-atom energy entries to zero, so their raw total energies omit the constant reference used during fitting. We restore this gauge without using the intermediate archive. For endpoint $j \in \{g, h\}$ and its own complete converged endpoint set $\mathcal{D}_j$, define

$$c_j = \frac{1}{|\mathcal{D}_j|} \sum_{k \in \mathcal{D}_j} \frac{E_k^{\text{DFT}} - E_{j,k}^{\text{raw}}}{N_k}, \quad \tilde{E}_{j,k} = E_{j,k}^{\text{raw}} + N_k c_j. \quad (9)$$

This is the least-squares per-atom intercept for the endpoint archive. The blend in Eq. (7) therefore interpolates the two endpoint-derived constants as well as the configuration-dependent energies. Energy gauges do not affect forces or any MD trajectory.

For K intermediate configurations and M Cartesian force components, the primary errors are

$$\text{MSE}_E(\lambda) = \frac{1}{K} \sum_{k=1}^K \left(\frac{E_{\lambda,k} - E_k^{\text{DFT}}}{N_k} \right)^2, \quad (10)$$

$$\text{MSE}_F(\lambda) = \frac{1}{M} \sum_{k,a} (F_{\lambda,k,a} - F_{k,a}^{\text{DFT}})^2. \quad (11)$$

The middle anchor minimizes the dimensionless, equally weighted objective

$$\lambda_M = \arg \min_{\lambda \in \{0, 0.025, \dots, 1\}} \frac{1}{2} \frac{\text{MSE}_E(\lambda)}{\text{MSE}_E(0)} + \frac{1}{2} \frac{\text{MSE}_F(\lambda)}{\text{MSE}_F(0)}. \quad (12)$$

Normalization prevents the units and relative number of energy and force observations from deciding the compromise. The smaller λ wins an exact tie. All grid points, the separate energy-only and force-only minimizers, RMSE and MAE, and errors stratified by configuration family are retained rather than reporting only the selected point.

The T_M archive was not used to fit either endpoint potential, so it is a held-out electronic temperature for the two ACE models. It is, however, consumed in full to identify the one scalar λ_M ; claims based on it are therefore calibration claims, not performance on an untouched test set. Uncertainty is quantified by paired configuration-level bootstrap resampling: each resample retains all force components

of a selected configuration, recomputes both methods' RMSE, and records their difference. Ten thousand resamples are drawn with fixed seed 15015, avoiding the false precision that would result from treating correlated Cartesian components as independent bootstrap units.

The primary energy label is the JSON field `Energy`, documented as the total DFT energy in the archive README and used by the supplied FitSNAP-readable dataset schema. The finite-temperature `FreeEng` field is processed through the same endpoint-only gauge and complete grid as a named thermodynamic-label sensitivity, but it cannot choose λ_M . This distinction matters because Mermin forces are associated with an electronic free-energy surface [Mermin, 1965], whereas the reproducibility question here requires the label actually used in the provided model workflow. No new DFT calculation is performed; the released data retain their VASP, PBE, `W_sv_GW` PAW, 700 eV plane-wave cutoff, consistent k -point meshes, and additional high-smearing bands [Kresse and Furthmüller, 1996, Kresse and Joubert, 1999, Perdew et al., 1996].

For the static comparator, the published family of sigmoids is evaluated without retuning:

$$\lambda_{\text{sig}}(T_e; T^*) = \left[1 + \exp\left(-\frac{T_e - T^*}{T^*/14}\right) \right]^{-1}, \quad (13)$$

$$T^* \in \{0.50, 0.75, 1.25, 2.50, 3.75, 5.00\} \text{ eV}.$$

The most aggressive member, $T^* = 0.50$ eV, is the primary accuracy comparator; all six are retained. CALM's static prediction at an electronic temperature is obtained by substituting $q(T_e)$ into Eqs. (7)–(8).

3.4 From local ionic temperature to an electronic proxy

The radiation-damage driver remains the center-of-mass-corrected local ionic temperature used by the layered baseline. For each atom i , LAMMPS `compute ave/sphere/atom` forms a neighborhood $\mathcal{N}_i$ containing the atom and all neighbors within $r_c = 5.28$ Å, computes the mass-weighted mean velocity $\bar{\mathbf{v}}_i$, and reports

$$T_i = \frac{\sum_{j \in \mathcal{N}_i} m_j |\mathbf{v}_j - \bar{\mathbf{v}}_i|^2}{3|\mathcal{N}_i|k_B}. \quad (14)$$

Equation (14) states the implemented LAMMPS convention, including its $3|\mathcal{N}_i|$ denominator. We express the result as a local kinetic-energy scale $\theta_i = k_B T_i$ in eV.

Equilibrium data set an explicit false-activation control. From 25 independent 1000 K BCC trajectories, each run for 10 ps and sampled every 0.1 ps after a 2 ps burn-in, we pool all θ_i and define

$$\theta_c = Q_{0.9999}(\theta_i). \quad (15)$$

This empirical cold guard targets at most one activation candidate per 10^4 pooled atom-frames under the calibration ensemble; it is not selected from a cascade outcome.

The caloric scale comes from exact-geometry matches between the two endpoint DFT archives. If $\mathcal{M}$ is the matched set, we use the robust configuration-wise coefficient

$$A = \text{median}_{k \in \mathcal{M}} \frac{(E_{H,k}^{\text{DFT}} - E_{L,k}^{\text{DFT}})/N_k}{T_H^2 - T_L^2}. \quad (16)$$

This construction uses the leading T_e^2 dependence suggested by the Sommerfeld expansion [Ackland, 2012]; it does not assume that a local ionic temperature equals the electronic temperature. The interquartile range of the configuration-wise coefficients is retained as a robustness diagnostic.

CALM maps the local kinetic scale to

$$\hat{T}_{e,i} = \min \left\{ T_H, \sqrt{T_L^2 + \frac{\alpha[\theta_i - \theta_c]_+}{A}} \right\}, \quad [z]_+ = \max(z, 0). \quad (17)$$

For $\theta_i \leq \theta_c$, Eq. (17) returns T_L and Eq. (6) gives exactly $\lambda_i = 0$. At the hot limit it returns T_H and gives exactly one. Although the positive-part and upper clipping in Eq. (17) are not themselves differentiable at their joins, the first two derivatives of q vanish at both electronic endpoints. Consequently their composition in Eq. (3) remains C^2 at activation and saturation.

201 The dimensionless α fixes aggressiveness, not accuracy. It is selected from the eight-point grid in
 202 Eq. (21) by minimizing absolute log mismatch in mean integrated excitation to the $T^* = 0.50$ eV
 203 sigmoid over five fixed orientations at each of 2 and 5 keV. Here

$$\eta(t) = \sum_i \lambda_i(t), \quad \chi = \int \eta(t) dt, \quad (18)$$

204 with the integral evaluated using the actual adaptive time increments. One log-linear interpolation
 205 between bracketing α values and one confirmation run are allowed. Neither instability labels, defect
 206 counts, nor thermal-spike volumes enter this selection. Matching χ controls aggregate exposure, so
 207 a comparison is not explained by a lower total χ alone. It does not equate spatial concentration or
 208 residence histories; whether the compact tails localize the matched exposure more effectively remains
 209 a dynamical hypothesis.

210 3.5 Nonconservative character and implementation

211 Force scaling by a configuration-dependent weight is generally nonconservative. Even for a single
 212 global $\lambda(\mathbf{R})$, differentiating the apparent mixture $\mathcal{U}_{\text{mix}} = (1 - \lambda)\mathcal{U}_g + \lambda\mathcal{U}_h$ gives

$$-\nabla_i \mathcal{U}_{\text{mix}} = (1 - \lambda)\mathbf{F}_{g,i} + \lambda\mathbf{F}_{h,i} - (\mathcal{U}_h - \mathcal{U}_g)\nabla_i \lambda. \quad (19)$$

213 Equation (2) omits the last term; atom-dependent weights introduce additional cross-derivative
 214 issues. CALM therefore does not claim to restore an energy Hamiltonian. Its compact plateaus and
 215 exposure-matched evaluation target unnecessary residence in the mixed region while retaining the
 216 exact force-level algebra under comparison.

217 The per-atom map is evaluated as LAMMPS atom-style variables and passed to `pair_style`
 218 `hybrid/scaled`; ACE forces are evaluated with PACE and the common short-range repulsion is
 219 overlaid once [Thompson et al., 2022, Wood et al., 2026]. All static predictions, selection tables, and
 220 configuration-level resamples are written before figure generation. OVITO performs the downstream
 221 structural and point-defect analyses [Stukowski, 2009, Larsen et al., 2016]; no such damage observable
 222 feeds back into Eqs. (12), (15), or (16).

223 4 Molecular-dynamics evaluation

224 4.1 Instantiated controller and exposure matching

225 During MD we evaluate Eqs. (6) and (17) directly; no fitted surrogate replaces their composition.

226 The static analysis fixes the intermediate anchor at $\lambda_M = 0.075$ and the caloric coefficient at
 227 $A = 1.3121123618 \text{ eV}^{-1}$.

228 The cold guard is calibrated in the same finite cell used for the dynamical campaign. Starting from the
 229 equilibrated configuration described below, we draw independent Maxwell-Boltzmann velocities for
 230 25 ground-ACE NVT trajectories at 1000 K. Each trajectory has a 2 ps burn-in and an 8 ps sampling
 231 interval; the local temperature in Eq. (14) is recorded every 0.1 ps. Applying the conservative
 232 “higher” empirical quantile to the pooled 2,073,600 atom-frame values gives $\theta_c = 0.1322868472 \text{ eV}$
 233 (1535.125 K). Thus the implemented map is

$$\lambda_i = q \left(\min \left\{ 0.5, \sqrt{0.001^2 + \frac{\alpha[k_B T_i - 0.1322868472]_+}{1.3121123618}} \right\} \right), \quad (20)$$

234 where all energy-like quantities are in eV and q is the compact C^2 map in Eq. (6). In particular, an
 235 atom at or below the guard has $\hat{T}_{e,i} = T_L$ and exactly $\lambda_i = 0$; numerical sigmoid tails are not used to
 236 approximate this plateau.

237 The remaining dimensionless coefficient α is set by excitation exposure, independently of any
 238 structural outcome. We enumerate

$$\mathcal{A} = \{0.01, 0.02, 0.04, 0.08, 0.16, 0.32, 0.64, 1.0\}. \quad (21)$$

239 For every $\alpha \in \mathcal{A}$, CALM is run on the same five orientations at both 2 and 5 keV. One shared
 240 $T^* = 0.50 \text{ eV}$ sigmoid series uses those ten matched cases, giving 80 CALM and 10 sigmoid

trajectories rather than duplicating the unchanged comparator eight times. If $\bar{\chi}_C(\alpha)$ and $\bar{\chi}_S$ denote the unweighted means over those ten trajectories, the selected grid point is

$$\alpha^* = \arg \min_{\alpha} \left| \log \frac{\bar{\chi}_C(\alpha)}{\bar{\chi}_S} \right|, \quad (22)$$

with a tie resolved toward the smaller α . Here χ is integrated only through the adaptive endpoint using Eq. (18); the actual, nonuniform physical times are used in trapezoidal quadrature. The calibration trajectories continue for 0.5 ps after that endpoint solely to retain the standard screening snapshots, and that interval is excluded from Eq. (22). Spike volume, instability, and final defects are not read during selection. The matched $T^* = 0.50$ eV sigmoid is the primary dynamical comparator, while $T^* \in \{0.75, 1.25, 2.50, 3.75, 5.00\}$ eV are unaltered secondary comparators; all use $\sigma = T^*/14$ and the same T_i .

4.2 Paired recoil protocol

We create periodic single-crystal BCC W at lattice constant $a = 3.176$ Å and equilibrate it for 100 ps at 1000 K and 1 atm with the ground-state ACE, a Nose-Hoover NPT integrator, and a 0.5 fs step. The resulting atomic state and its unrecoiled reference lattice are held fixed across methods. A deterministic atom nearest the box center is the primary knock-on atom (PKA), and its equilibrated velocity is replaced by the velocity corresponding to the requested recoil energy and direction. Directions are isotropic draws from a PCG64 stream with seed 150150. The same direction, equilibrated-data hash, and PKA identifier define a pair across methods.

The declared recoil grid is

$$E_{\text{PKA}} \in \{1, 2, 5, 10, 15, 20, 30, 50\} \text{ keV}, \quad (23)$$

with 25 orientations per energy. Primary trajectories use CALM, the exposure-matched $T^* = 0.50$ eV sigmoid, ground-state ACE, and the supplied Marinica EAM potential [Marinica et al., 2013]; the other five published sigmoids form secondary series. CALM and sigmoid trajectories blend both immutable PACE models. The ground-ACE control uses only the ground model. All three use the same unscaled ZBL overlay with inner and outer radii 3.927 and 4.488 Å, respectively. The supplied EAM is used as a standalone potential, without an additional ZBL term. Production ACE evaluation uses PACE’s recursive evaluator. A separate 128-atom trajectory check against the product evaluator used in the supplied example bounded the largest recorded state-component difference by 9.96×10^{-13} ; this is an implementation check, not cascade evidence.

During the ballistic phase, atoms in a static shell of thickness $2a$ from any of the six faces are coupled to a 1000 K Nose-Hoover thermostat with 0.1 ps damping, while the complementary interior is NVE. Group membership is fixed at the initial geometry, as in the supplied LAMMPS construction. The PKA is required to belong to the interior. LAMMPS `fix dt/reset` updates every 100 steps, constrains the largest displacement per step to 0.05 Å, and permits steps from 10^{-6} to 0.5 fs. After the step first returns to 0.5 fs, a second 100-step block must remain at that value before the adaptive phase ends. The entire cell is then thermostatted at 1000 K for 30 ps at 0.5 fs. Atomic snapshots are written at the adaptive endpoint, 0.5 ps later, and 30 ps later. Lost atoms, an exhausted adaptive loop, nonfinite values, or a missing completion marker are explicit failures.

The force-layered trajectories have no reported blended potential energy or pressure. For atom-style scale factors, LAMMPS `hybrid/scaled` supplies forces but reports zero energy and virial for the variably scaled substyles; interpreting those quantities as a mixed Hamiltonian would also contradict Eq. (19). We use kinematic and structural observables instead. At every tenth MD step the harness records physical time, the current step size, $\eta = \sum_i \lambda_i$, $\max_i \lambda_i$, and populations above $\lambda = 0.1, 0.5$, and 0.9. Duplicate records at LAMMPS run boundaries are replaced by the last record at that time before trapezoidal integration.

4.3 A declared finite-volume deviation

The original design called for containment pilots over $n \in \{16, 20, 24, 28, 32, 40\}$ conventional cells per side, selecting the smallest box whose disordered region did not meet the thermostat and whose shell remained below 1500 K. Those pilots were set aside before any production outcome was inspected: timing measurements on the available 32-CPU node showed that the resulting 25-member,

289 nine-method campaign could not be completed at even the smallest registered size. The executable
290 MD tier therefore uses one equilibrated $n = 8$ box, containing 1,024 atoms with a nominal side of
291 25.408 Å, for every method, energy, and orientation.

292 This is a compute-bounded controller stress test, not a hidden approximation to the intended geometry.
293 Its nominal volume is about 250 times smaller than the smallest 250,000-atom cell in the published
294 cascade study, whose cells grew further with recoil energy [Wood et al., 2026]. Moreover, the
295 published face thermostat occupied 2.5% of the cell length, whereas the registered implementation
296 here uses a fixed two-lattice-constant shell. In the $n = 8$ box that shell leaves an interior only
297 four conventional cells wide. Periodic-image interaction, thermostat overlap, and loss of spatially
298 separated subcascades are therefore expected geometric effects, especially at the upper end of Eq. (23).
299 Pairing holds those effects fixed when contrasting controllers, but cannot turn the contrasts into bulk
300 or cell-size-converged recoil thresholds.

301 4.4 Checkpointed execution

302 Each expensive trajectory is divided without altering its equations of motion. Stage 1 is the literal
303 trajectory prefix through the adaptive phase and first 0.5 ps of whole-cell NVT annealing. It writes
304 a binary restart while the annealing fix is active. Stage 2 reads that exact atom state, reconstructs
305 the identical map and pair styles, and reuses the same fix identifier so that LAMMPS restores the
306 serialized Nose-Hoover chain before integrating the remaining 29.5 ps. The same executable and
307 MPI rank count are required, the restart hash is checked, and the two $\eta(t)$ fragments are merged by
308 physical time with their common boundary retained once.

309 We validated this decomposition independently in a deterministic 128-atom CALM smoke trajectory.
310 One-shot and staged runs had zero recorded difference in final positions, velocities, forces, and
311 per-atom λ , as well as in every unique η record and integrated χ ; the comparison tolerance was
312 2×10^{-12} . This establishes equivalence of the execution paths only. It does not validate the potential,
313 the finite cell, or any damage trend.

314 A selected continuation that exits nonzero is replayed once with the identical rendered input, binary
315 checkpoint, LAMMPS executable, and eight-rank MPI layout. Both attempts are retained separately.
316 A reproduced failure is a terminal 30 ps survival outcome: its independently completed adaptive
317 and spike observables remain valid, but no final Wigner–Seitz count is imputed. Consequently,
318 final-dump means and paired contrasts use exactly the reported complete-case denominator, while
319 completion/selection is reported as a separate survival fraction.

320 CALM and sigmoid switching series advance in increasing recoil energy. A switching series
321 terminates at the first energy with at least 7 unstable trajectories among exactly 25 valid volume
322 comparisons, operationalizing the published “greater than 25%” rule; the triggering energy remains
323 in the series. Rendered higher-energy cases are identified as outside the sequential series and are not
324 used to revise the trigger. An incomplete lower-energy denominator makes subsequent termination
325 status unevaluable rather than treating missing cases as stable. Stage 1 supplies the complete stability
326 screen before eligible trajectories are continued through stage 2. The nonswitching ground-ACE and
327 EAM controls run through every declared recoil energy and are not truncated by this switching-series
328 rule.

329 4.5 Structural observables

330 All structural analyses use OVITO 3.13.1 [Stukowski, 2009]. At both the adaptive endpoint and
331 +0.5 ps, Polyhedral Template Matching (PTM) is applied with RMSD cutoff 0.1 [Larsen et al.,
332 2016]. Atoms whose PTM structure is not BCC are clustered with a 3.2 Å cutoff and periodic
333 unwrapping. This cutoff, which includes the first two BCC neighbor shells, is explicit because the
334 publication did not specify its connectivity rule. If the PKA is non-BCC we select its connected
335 component; otherwise we select the component containing the nearest non-BCC atom under the
336 periodic minimum-image convention, with particle identifier breaking ties. The fallback identity and
337 distance are logged. An empty non-BCC set has zero volume. A selected set with fewer than four
338 atoms is a separately marked, geometrically degenerate zero.

339 The first real prefix exposed an analysis-geometry limitation: OVITO cannot construct a 10 Å alpha
340 shape in a periodic cell only 25.52 Å wide. After periodic cluster unwrapping, we therefore copy

exactly the selected coordinates into a nonperiodic axis-aligned bounding cell, translate them rigidly, and pad each face by 20 Å (twice the probe radius). No atom is added, removed, or rescaled. OVITO’s alpha-shape construction then uses the registered 10 Å (1 nm) probe, no smoothing, and reports the filled volume. Translation and periodic-wrap/unwrapping tests changed the computed volume by at most $1.71 \times 10^{-13} \text{ Å}^3$ (tolerance 10^{-9} Å^3). This establishes invariance of the repaired coordinate construction, not equivalence to an alpha shape in the unavailable original periodic geometry. The analysis-cell replacement is an explicit consequence of the finite-volume deviation in Section 4.3. The two operational spike volumes are

$$V_0 = V(t_{\text{adaptive}}), \quad V_{0.5} = V(t_{\text{adaptive}} + 0.5 \text{ ps}), \quad (24)$$

and instability is the strict comparison $V_{0.5} > V_0$, without a numerical tolerance. A Delaunay failure after this repair is an analysis failure, never coerced to zero. Total non-BCC population and cluster count accompany the single-component volume so that detached subcascades are visible.

The 30 ps snapshot is compared with the stored, unrecoiled equilibrated reference by OVITO Wigner-Seitz analysis. A PKA gives the small periodic box net momentum, so a healed lattice can translate relative to the reference. Before assigning occupancies, we match particle identifiers and subtract only the equal-mass center-of-mass translation

$$\Delta \mathbf{r}_{\text{COM}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{r}_i^{\text{unwrap}} - \mathbf{r}_{i,0}^{\text{unwrap}}). \quad (25)$$

Every atom is W, and the recorded image flags define the unwrapped positions. No rotation, strain, or internal displacement is removed; unregistered counts and $\|\Delta \mathbf{r}_{\text{COM}}\|$ remain in each case record. This repair was fixed after the first four completed continuations exposed rigid-drift sensitivity and before any damage aggregate was computed. Affine cell remapping is disabled and the periodic minimum-image convention is enabled. Current and reference particle counts, identifier sets, particle types, and periodic flags must agree, and their cell matrices must match to 10^{-8} Å . We require equal vacancy and interstitial counts and report

$$N_{\text{FP}} = \frac{N_{\text{vac}} + N_{\text{int}}}{2}. \quad (26)$$

A mismatch is retained as an explicit analysis failure rather than rounded.

4.6 Statistical analysis and external reference curves

The campaign manifest, rather than the set of successful output files, fixes the denominator. For each method and energy we report every trajectory, stage-specific valid count, arithmetic mean, sample standard deviation, and a 95% nonparametric percentile interval for the replicate mean from 10,000 resamples. The base seed is 150150; a SHA-256-derived seed makes every table row independently reproducible. Paired contrasts are formed only when energy, replicate, orientation, and equilibrated-data hash agree. They are reported as left-minus-right means with sample standard deviations and paired bootstrap intervals. Missing or malformed values are never imputed as zero, although an independently successful stage of an otherwise failed case may contribute its own observable. Bootstrap intervals are conditional on the reported valid cases and do not model the missingness or failure process.

The primary dynamical contrast is CALM against the exposure-matched $T^* = 0.50 \text{ eV}$ sigmoid. Ground ACE tests whether a stable controller has merely collapsed to cold dynamics, and Marinica EAM provides a classical potential reference. The prespecified finite-box stability condition requires fewer than 7 CALM instabilities at an energy where the matched sigmoid reaches 7 of 25, together with mean adaptive χ within 10%. Preservation of the Sommerfeld damage direction requires CALM mean N_{FP} and V_0 to be below ground ACE at a majority of non-triggering energies. These are directional ensemble criteria, not tests chosen from a favorable confidence interval. The implemented diagnostics comprise offline activation statistics on the ground-ACE equilibrium guard ensemble and aggregate η , maximum- λ , and threshold-population traces. They do not measure a CALM-versus-ground dynamical drift, continuous per-atom residence histories, or endpoint-force work, so no inference about those quantities is made.

For large-cell context only, we digitized the plotted markers in Figures 4–7 of the supplied article [Wood et al., 2026]. Axis calibration was performed on the embedded 1,347-by-1,000-pixel raster;

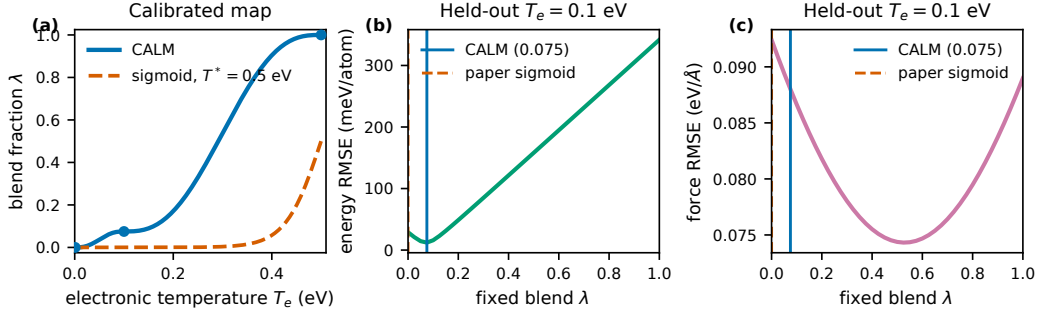


Figure 1: **Accuracy calibration.** (a) CALM’s endpoint-flat electronic map and the most aggressive published sigmoid. Markers are the three supplied electronic-temperature anchors. (b,c) Configuration-balanced energy and component-balanced force RMSE on all 2444 intermediate configurations for every fixed blend on the declared grid. Vertical lines locate the selected CALM middle anchor and the value produced by the $T^* = 0.5$ eV sigmoid at $T_e = 0.1$ eV.

occluded markers were reconstructed from visible edges and flagged. Typical reading uncertainty is 1–2% for logarithmic quantities, 1–2 nm³ for the plotted spike-volume differences, and 0.2 Frenkel pairs for unobscured difference markers, with larger uncertainty assigned to occlusions and high counts. These values are labelled *source-figure digitizations*: they are neither author source data nor observations in our bootstrap, and the source plots provide no replicate-level spread. Apparent inconsistencies between source prose and markers are retained rather than resolved by deleting points.

5 Intermediate-state accuracy

The immutable endpoint potentials were evaluated on every converged record in the three delivered archives: 9,467 configurations at T_L , 2,444 at T_M , and 9,445 at T_H . The intermediate archive contains 29384 atoms (88,152 force components). All LAMMPS evaluations completed; the failure logs are empty. The endpoint-only energy gauges are -6.109206 and -6.340328 eV/atom for $\mathcal{U}_g$ and $\mathcal{U}_h$, respectively. They repair only the omitted atomic reference and never enter an MD force.

Figure 1 shows the complete preregistered blend sweep. The energy-only minimum is $\lambda = 0.075$, whereas the component-balanced force minimum is $\lambda = 0.525$. This separation is direct evidence that the intermediate surface is not obtained by a single linear progression that is simultaneously optimal for both observables. The normalized joint criterion selection selects $\lambda_M = 0.075$; no cascade observable was available to that selection. The resulting two-piece map passes through $(T_L, 0)$, (T_M, λ_M) , and $(T_H, 1)$ with compact C^2 tails (Fig. 1a).

Figure 2 evaluates those maps, rather than a freely reoptimized blend, at every delivered electronic-temperature archive. The two controllers are nearly coincident at the cold endpoint, whereas CALM’s calibrated middle anchor improves both observables at T_M and its exact hot plateau recovers the supplied high-temperature potential at T_H . The sigmoid has $\lambda = 1/2$ at $T_H = T^*$ and therefore does not recover that endpoint. This three-temperature comparison tests interpolation and endpoint behavior only; the endpoint archives trained the two immutable ACE models and are not held-out accuracy sets.

At the intermediate temperature, CALM reduces energy RMSE from 29.02 to 12.16 meV/atom and force RMSE from 0.0924 to 0.0880 eV/Å (Table 1). These are reductions of 58.1% and 4.8%, respectively. Conditional on the selected λ_M , in 10,000 paired configuration bootstraps the 95% interval for CALM-minus-sigmoid energy RMSE is $[-17.37, -16.32]$ meV/atom and that for force RMSE is $[-0.0049, -0.0039]$ eV/Å; neither interval touches zero. The comparison is deliberately against the most aggressive published sigmoid: at $T_e = 0.1$ eV it gives $\lambda = 1.3674 \times 10^{-5}$, and every larger published T^* is closer still to the ground endpoint.

The available source article does not contain the supplemental intermediate-temperature blend figure, so its energy zero, weighting, and configuration subset cannot be reconstructed. We therefore

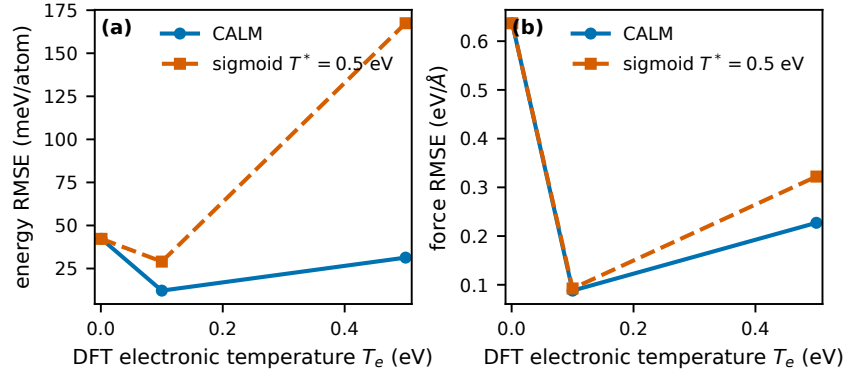


Figure 2: **Errors obtained by applying each complete map.** Energy (a) and force (b) RMSE are shown for all converged configurations in each delivered archive. Only $T_e = 0.1$ eV is withheld from both endpoint fits; the endpoint values diagnose the required limits.

Table 1: All-configuration errors at the intermediate electronic temperature. Parentheses give paired-bootstrap 95% intervals for the CALM-minus-sigmoid RMSE difference.

map at $T_e = 0.1$ eV	energy RMSE (meV/atom)	force RMSE (eV/Å)
sigmoid, $T^* = 0.5$ eV	29.02	0.0924
CALM ($\lambda_M = 0.075$)	12.16	0.0880
paired difference	-16.86 ([-17.37, -16.32])	-0.0044 ([-0.0049, -0.0039])

retained the prespecified all-archive analysis rather than adjusting it to an unavailable result. Two fully enumerated sensitivities illustrate the dependence: using VASP FreeEng rather than the README-documented Energy moves the all-data energy optimum to 0.225, and excluding the ObjectiveFunctions and Disordered_Struc families moves the force optimum to 0.350. The primary all-data result remains the one fixed before the sweep was inspected. These shifts offer possible reconciliations, but do not identify the unpublished analysis protocol. More generally, a scalar blend is not an intrinsic material constant until the thermodynamic label, energy gauge, structure distribution, and error measure have all been specified.

Finally, exact geometry matching between endpoint archives yields 8793 structures. Their median coefficient in Eq. (16) is 1.3121 eV^{-1} , with interquartile interval $[1.2887, 1.3385] \text{ eV}^{-1}$. The narrow central interval shows that one coefficient represents the middle half of the matched structures under the chosen two-endpoint T_e^2 parameterization. Because each coefficient uses only two temperatures, this result does not validate quadratic temperature dependence. The logged rejected matches and full range further prevent the interpolation from being mistaken for a universal electronic equation of state.

6 Controller calibration and equilibrium activation

All 90 exposure-calibration trajectories completed without a LAMMPS failure. The adaptive-phase exposure increased monotonically across the eight-point α grid. Equation (22) selected $\alpha^* = 0.320$ without invoking the allowed interpolation step: CALM gave a pooled mean $\chi = 50.637$ atom ps, compared with 47.812 atom ps for the $T^* = 0.50$ eV sigmoid, a signed mismatch of +5.9%. At 2 keV the corresponding means (sample standard deviations) were 42.007 (3.129) and 39.314 (3.326) atom ps; at 5 keV they were 59.266 (11.596) and 56.310 (11.729) atom ps. Thus the frozen CALM controller meets the prospective 10% aggregate-exposure tolerance. This calibration controls total adaptive exposure; it does not assert equality of its spatial or temporal distribution.

The equilibrium guard calibration ensemble contains 2073600 local atom-frame temperatures. Evaluating the frozen controller on that ensemble gives median $\lambda = 0$, mean 1.19×10^{-6} , and maximum 0.061; only 0.0100% of atom-frames are nonzero and none exceeds $\lambda = 0.1$. The matched sigmoid is nonzero on every sampled atom-frame and has mean 9.11×10^{-6} . These results demonstrate the

Table 2: Headline results. Cascade blocks contain 25 orientations; 30 ps observables use completed trajectories and the survival row reports completed/selected counts. The comparator is the reproduced $T^* = 0.50$ eV sigmoid unless stated otherwise.

Axis	Quantity	CALM	Comparator/context
Held-out accuracy	Energy RMSE (meV atom ⁻¹)	12.161	29.016
Held-out accuracy	Force RMSE (eV Å ⁻¹)	0.088008	0.092415
Finite-volume stability	Highest included / sequential outcome	10/ incomplete at 15 keV	15/ incomplete at 20 keV
30 ps survival	Completed / selected trajectories	95/100	114/125
Published large-cell context	Reported highest recoil energy	CALM was not in the source study	$T^* = 0.50$ eV: 15 keV; $T^* = 5.0$ eV: 200 keV
Exposure match	Max $ \Delta\chi $ over common energies	10.234% (4 energies)	calibration: 5.9076%
Surviving defects	CALM below ground ACE	1/4 energies	Δ range -1.0952 to 0.92 pairs; paired $n=21$ to 25
Peak spike volume	CALM below ground ACE	3/4 energies	Δ range -0.93196 to 0.017709 nm ³

intended empirical false-activation control and exact cold plateau. Because the temperature samples came from ground-ACE trajectories, this offline comparison does not measure dynamical energy drift under CALM, and no such drift claim is made.

7 Finite-volume recoil dynamics

7.1 The registered stability range does not extend

Figure 3 reports every evaluable switching-series prefix. CALM's last complete energy is 10 keV and its sequential outcome is incomplete at 15 keV. The exposure-matched $T^* = 0.50$ eV sigmoid reaches 15 keV, with outcome incomplete at 20 keV. The other reproduced sigmoid series give, respectively, $T^* = 0.75$ eV: 10 keV and incomplete at 15 keV; $T^* = 1.25$ eV: 20 keV and incomplete at 30 keV; $T^* = 2.50$ eV: 30 keV and incomplete at 50 keV; $T^* = 3.75$ eV: 50 keV and not triggered through 50 keV; and $T^* = 5.0$ eV: 50 keV and triggered at 50 keV. Open markers retain observed partial-denominator fractions for diagnosis but are not 25-case stopping statistics.

At 15 keV, CALM produced 24 valid volume comparisons, of which 5 increased strictly. Replicate 20 reached the adaptive endpoint and then lost two atoms during the fixed-step probe; an identical-input retry reproduced the same failure. The $T^* = 0.75$ eV sigmoid likewise supplied only 24 valid comparisons, with 3 strict increases; its missing case remained in a million-kelvin adaptive runaway until operator termination. Neither missing trajectory is promoted to a volume increase. Consequently, the registered fixed-denominator stability-extension condition is not met: CALM has no evaluable energy beyond 10 keV, and no favorable outcome at the first incomplete energy can be inferred from the observed subset.

The matched sigmoid subsequently became incomplete at 20 keV. It yielded 22 valid comparisons and 7 strict increases. One trajectory lost two atoms during the fixed-step probe and reproduced that failure under an identical forced retry; two more remained in million-kelvin adaptive runaways until they met the disclosed resource ceiling. Although the observed numerator equals the nominal stopping count, it is seven of 22, not seven of 25, and is therefore recorded only as a failed denominator. This second failure boundary reinforces the central negative result: compact tails and matched integrated exposure did not make either aggressive controller evaluable across the declared sweep in this small cell.

A third boundary occurred for the $T^* = 1.25$ eV sigmoid at 30 keV: 24 comparisons were valid and 7 increased. The missing orientation lost one atom during the fixed-step probe, and its complete thermo trace and error reproduced under an identical retry. This repeated sequence of exact-denominator

Finite-volume cascade ensemble — means use the shown valid-replicate denominators

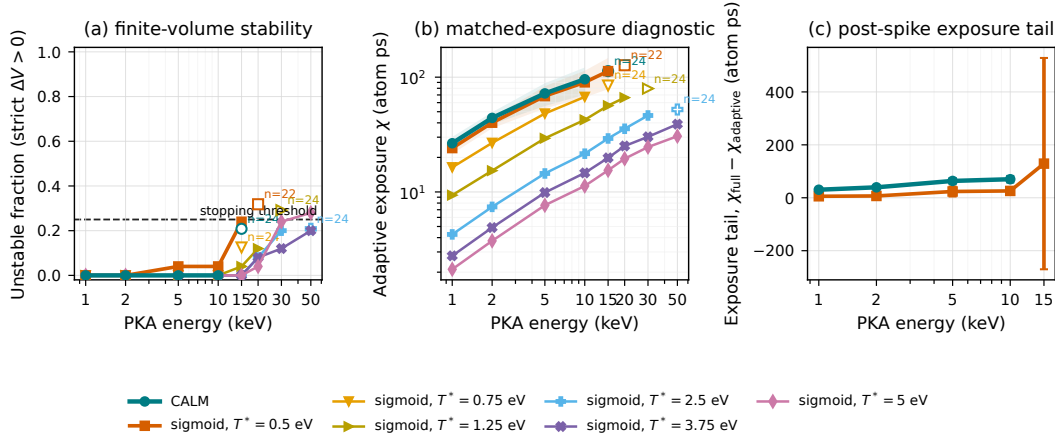


Figure 3: **Finite-volume switching behavior.** (a) Strict thermal-spike volume increases after the adaptive phase; the horizontal line is the published greater-than-25% stopping rule, operationalized as 7 of 25. Open points show incomplete denominators and are not used to trigger or clear a series. (b) Adaptive excitation exposure for CALM and all six reproduced sigmoids. (c) Post-spike exposure after the adaptive endpoint for the primary exposure-matched pair. Lines connect only valid sequential prefixes; error bars are sample standard deviations.

failures is itself informative: the nominal volume numerators near each boundary cannot rescue a series whose trajectories cease to be executable.

7.2 Thirty-picosecond survival exposes a second failure mode

The registered stopping rule uses the two stage-1 spike snapshots, but passing that screen does not guarantee a complete anneal. Of 1450 selected continuations, 1432 reach 30 ps and 18 lose atoms before the final snapshot. CALM completes 95/100 selected trajectories; the exposure-matched sigmoid completes 114/125. Failures also occur for the $T^* = 0.75$ eV sigmoid, while every selected trajectory for the remaining controllers and both controls completes.

Every failed continuation was archived and replayed with its original binary checkpoint, rendered input, executable, and MPI layout. All 18 exact replays fail again; the retained excitation trace and terminal atom-loss signature are identical within each pair of attempts. The CALM and matched-sigmoid failure sets at 10 keV also contain the same orientations. This orientation correspondence and exact replay exclude a launcher transient, but the small periodic box prevents a unique attribution to controller shape, force nonconservation, or image interaction. We therefore report completion as a separate survival outcome and never substitute a defect count for a missing final snapshot.

7.3 Matched exposure conceals a different state distribution

Across the 4 complete adaptive-energy groups shared by CALM and the matched sigmoid, the largest absolute mean-exposure mismatch was 10.234%. Thus the pooled calibration mismatch of 5.9% does not imply exact energy-by-energy matching in production. Moreover, the reproduced matched-sigmoid exposure exceeds the approximately digitized source-paper exposure by factors ranging from 2.6624 to 18.687 over 5 common energies. This scale gap is consistent with the deliberately small, thermostat-dominated box and is why the source curve and reproduced curve are shown side by side rather than pooled.

At the highest common energy, 10 keV, CALM and the matched sigmoid have mean peak η values of 725.11 and 718.43 atoms. Their time-averaged populations above $\lambda = 0.1$ are 698.26 and 541.62 atoms; above 0.5 they are 495.74 and 483.41; and above 0.9 they are 338.04 and 433.73. The corresponding post-adaptive exposure tails are 70.067 and 25.85 atom ps. These measured populations

Finite-volume cascade damage — filled solid: reproduced mean $\pm$ SD; open dotted: approximate paper digitization

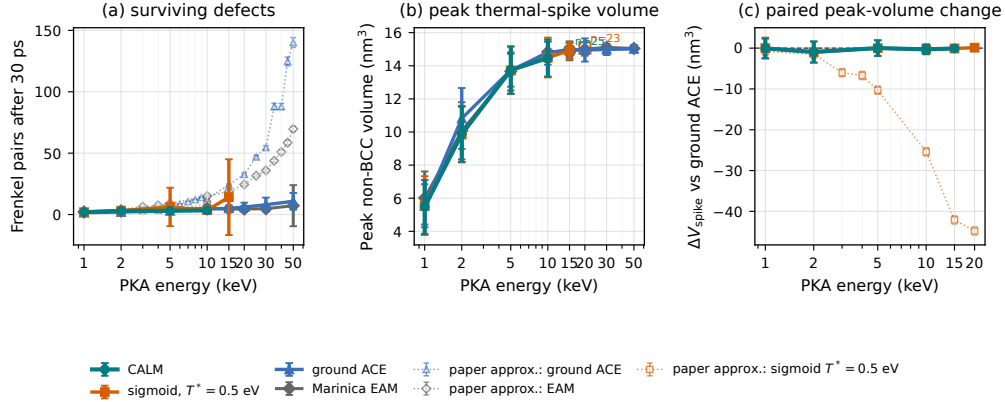


Figure 4: **Damage observables and the two baselines.** (a) Surviving Frenkel pairs after 30 ps, (b) peak connected non-BCC volume, and (c) the paired peak-volume difference from ground ACE. Filled solid curves are the 1,024-atom reproduction (mean and sample standard deviation over the shown valid denominator). Open dotted curves are approximate digitizations of the published large-cell figures and carry their reading uncertainty; they are contextual, not pooled observations.

test the compact-tail hypothesis directly at aggregate level. They do not identify continuous per-atom residence or mixed-force work, neither of which was recorded.

7.4 The two damage directions do not jointly persist

Figure 4 compares all completed 30 ps outcomes. Relative to paired ground ACE, CALM has fewer mean surviving Frenkel pairs at 1 of 4 common energies; its paired mean difference ranges from -1.0952 to 0.92 pairs, using paired complete-case counts from 21 to 25. CALM has a smaller peak non-BCC volume at 3 of 4 energies, with a paired difference range of -0.93196 to 0.017709 nm³. The spike-volume direction therefore meets its majority component, whereas the surviving-defect direction does not. The prespecified joint preservation criterion fails. These direction counts use every eligible energy without selecting by confidence interval; the Frenkel means remain conditional on survival where a continuation failed.

The source study reports its $T^* = 0.50$ eV sigmoid through 15 keV and its $T^* = 5.0$ eV sigmoid through 200 keV in much larger cells. The present sequential outcomes and much larger reproduced χ values therefore do not replicate those large-cell thresholds. The paired finite-volume damage contrasts remain useful as a controller stress test, but their agreement or disagreement with the digitized source direction cannot remove the geometry confound.

8 Discussion

8.1 What the controller changes—and what it does not

CALM turns an underdetermined sigmoid into three separable decisions. The intermediate electronic-state mixture is fixed by the supplied finite-temperature DFT archive; the equilibrium guard is fixed by a false-activation quantile; and dynamical aggressiveness is fixed by exposure to a declared comparator. The compact minimum-jerk map then supplies exact cold and hot plateaus without changing either ACE model or the force-layering algebra. This separation is useful even when a measured dynamical contrast is null: it identifies which empirical evidence supports each part of the map, and prevents a recoil outcome from retrospectively choosing an ionic switching threshold.

The construction does not make the layered dynamics conservative. As Eq. (19) shows, the implemented force omits terms that would arise from differentiating a coordinate-dependent mixture, and atom-specific weights create cross-coordinate consistency conditions beyond the global illustration.

Exact plateaus can reduce the portion of phase space on which this inconsistency acts, but they cannot cancel the missing derivatives while $0 < \lambda_i < 1$. Consequently, any difference between CALM and an exposure-matched sigmoid should be interpreted as a consequence of where and when mixed forces are applied, not as recovery of a conserved electronic free energy. This is also why no “blended potential energy” or pressure is inferred from the variable-scaled LAMMPS pair style.

The finite-volume experiment supplies a direct counterexample to the stronger compact-tail hypothesis. CALM becomes incomplete one energy before the matched sigmoid even though its calibration exposure differs by only 5.9% and its equilibrium tail is essentially absent. The failure is not a marginal volume classification: one trajectory loses atoms at the same step on an identical retry. The small periodic cell prevents us from assigning that divergence uniquely to the controller, the force blend, or a cascade interacting with its images, but it does rule out a stability claim for the realized protocol. Treating the remaining 24 orientations as a complete ensemble would reverse the logic of the registered denominator.

The checkpoint continuations reveal a separate survival problem after that registered screen. CALM completes only 95/ 100 selected 30 ps trajectories, compared with 114/ 125 for the matched sigmoid. All 18 failures reproduce under exact replay. Because a failure can depend on the same hot trajectory that would otherwise determine damage, the missing final configurations are not plausibly missing at random; complete-case Frenkel means must be read jointly with survival rather than as unbiased replacements for the planned 25-case means.

Exposure matching narrows, but does not eliminate, that interpretation. Two controllers with the same integrated χ may differ in peak population, spatial concentration, residence-time distribution, and correlation between λ_i and endpoint force disagreement. Those differences are precisely the hypothesized value of a compact guard, but χ alone cannot identify the mechanism. The default confirmatory trace records η , maximum λ , and three threshold populations; it does not record a full per-atom residence history or the endpoint-force work integral. We therefore do not infer those unmeasured quantities from a favorable volume trajectory. A definitive work accounting would require paired endpoint-force reruns, and residence statistics would require enabling the much larger per-atom trace.

8.2 A scalar blend depends on the target distribution

The intermediate-temperature analysis exposes a more general limitation of two-surface interpolation: there is no label-independent “correct” scalar blend. Energy and force residuals weight different distortions of the potential-energy surface, and their optima need not coincide. The optimum can also move when configuration families are reweighted or when the VASP `Energy` field is replaced by `FreeEng`. We use `Energy` for the primary analysis because the archive README documents it as the reference total-energy field, and enumerate `FreeEng` only as a thermodynamic-label sensitivity. Mermin free energy remains the natural finite-electronic-temperature surface for force generation [Mermin, 1965]; that physical observation is not license to replace the documented target after seeing a sweep.

This distinction also explains why our all-archive curve should not be called an exact reproduction of an externally quoted optimum. The delivered article does not contain the cited supplementary blend curve, and it does not identify the configuration subset, category weights, or energy-zero convention behind that value. Its aggregate endpoint errors are likewise not all recovered by uniformly weighting every delivered configuration and force component. By retaining the primary label, the complete archive, both endpoint-derived energy gauges, and explicit label/subset sensitivities, we obtain a reproducible calibration for this study without reverse-engineering an unavailable analysis. The resulting λ_M is therefore conditional on the declared loss and archive distribution, rather than a universal material constant.

8.3 Finite-volume scope of the cascade evaluation

The dominant limitation is geometric, not statistical. The 1,024-atom box in Section 4.3 is far smaller than both the registered containment candidates and the published 250,000–16,000,000-atom cells [Wood et al., 2026]. At this scale a recoil can meet its periodic image, the disordered region can encounter the thermostat before a bulk-like thermal spike forms, and the fixed $2a$ shell occupies a large fraction of the box. The 1 nm surface-mesh probe is itself comparable with the box half-width.

These are systematic changes to heat removal, recombination, and cluster topology; increasing the number of orientations cannot average them away. Net PKA momentum also translates the entire small lattice appreciably during the anneal. Equation (25) removes that rigid coordinate mode before Wigner-Seitz assignment, but the logged correction is another finite-volume diagnostic rather than evidence of bulk-like momentum removal.

Pairing nevertheless supports a narrower inference. Because CALM, every sigmoid, and ground ACE share the same initial coordinates, orientation, PKA, box, shell, and integration schedule, their within-pair differences test how the controller behaves under one severe finite-volume stress. Pairing does not guarantee that controller and geometry interact additively. In particular, a map that appears favorable when a thermostat is only a few lattice constants from the PKA need not retain that ordering when electronic excitation and lattice heat propagate through a large interior. The finite-box termination energy is thus neither a bulk stability limit nor a reproduction of a published recoil threshold. Absolute spike volumes and Frenkel-pair counts should not be overlaid as though the two cell scales sampled the same physical system.

The same caution applies to α . It is exposure-matched in the $n = 8$ cell, and $\chi = \int \sum_i \lambda_i dt$ is extensive. CALM's exact cold plateau and a sigmoid's small equilibrium tail scale differently with atom count, so transfer of the fitted α to a much larger box is a testable hypothesis rather than a consequence of Eq. (22). A large-cell study should repeat the exposure calibration or match an explicitly background-subtracted intensive exposure, then perform the registered containment check before interpreting damage. Alpha-shape failures in the present small periodic cell are themselves geometry diagnostics and are not valid zero-volume observations.

8.4 Relation to electronic-energy models

CALM occupies a deliberately narrower layer of the modeling hierarchy than a two-temperature model. Two-temperature MD evolves an electronic energy field and represents stopping, diffusion, electron-phonon coupling, and return of energy to the ions [Duffy and Rutherford, 2007, Zarkadoula et al., 2015, Sand et al., 2025]. CALM does none of these: its $\hat{T}_e$ is a local caloric proxy driven by ionic kinematics, with no electronic conservation law or transport equation. It selects between two already trained electronic-temperature force surfaces. The approaches could ultimately be coupled, but the state variable and energy exchange would then need to be made consistent to avoid counting the same dissipation twice [Zhang and Weber, 2020].

Conservative adaptive-precision potentials offer another important contrast. By defining switching at the energy level and retaining derivative terms, they can preserve an underlying Hamiltonian within their construction [Immel et al., 2025]. CALM instead preserves the general per-atom hybrid/scaled mechanism of layered Sommerfeld ACE [Wood et al., 2026, Thompson et al., 2022]. Its contribution is an auditable state map for that existing approximation, not a substitute for a conservative model. The appropriate next test is therefore twofold: repeat the paired controller study in containment-converged cells, and compare against a formulation in which the missing switching derivatives or a dynamically evolved electronic state are present.

Finally, the digitized source curves provide historical scale and a check on qualitative direction, but not independent raw observations. Their raster uncertainty, reconstructed occlusions, missing replicate spread, and an internal mismatch between the plotted and prose-reported endpoint of the most aggressive sigmoid preclude formal pooling with our trajectories. Any claim based on the recoil trajectories is accordingly bounded to paired finite-volume dynamics. The completed experiment establishes a negative stability result, a reproducible continuation-survival failure, and split damage directions in that box: peak spike volume is lower at a majority of CALM energies, but surviving defects are not. Thus the prespecified physical- preservation condition also fails. These outcomes accompany, rather than erase, the immutable-potential DFT calibration, exposure match, and equilibrium guard. Cell-size-converged extension of the stable recoil range, bulk defect suppression, and a residence/work mechanism remain claims for a larger calculation, not conclusions manufactured from the small-cell evidence.

9 Conclusion

CALM replaces a free sigmoid threshold with an auditable sequence of calibrations while leaving both endpoint ACE models and their force-layering implementation unchanged. On the independently supplied intermediate-temperature archive, the normalized energy-force criterion selected $\lambda_M = 0.075$. Relative to the most aggressive published sigmoid, this map reduced energy RMSE from 29.02 to 12.16 meV/atom and force RMSE from 0.0924 to 0.0880 eV/Å. The compact C^2 construction reaches exact endpoint plateaus, and its quantile guard limited activation to 0.0100% of the sampled ground-ACE equilibrium atom-frames. Independently, exposure matching fixed $\alpha = 0.320$ within 5.9% of the declared sigmoid target.

The recoil test rejects the stronger hypothesis. CALM's exact denominator fails at 15 keV, whereas the matched sigmoid's fails at 20 keV; neither failure is relabelled as a spike-volume increase. Equal integrated exposure also conceals markedly different threshold populations at 10 keV, so χ is a useful fairness constraint but not a sufficient stability coordinate. The continuations reinforce that result: 1432 of 1450 selected trajectories reach 30 ps, and all 18 failures reproduce under exact replay. Over CALM's four selected energies, it yields fewer surviving Frenkel pairs than ground ACE at only 1/4 energies, although peak disordered volume is smaller at 3/4. The joint physical-preservation criterion therefore fails. Those conditional paired directions describe a 1,024-atom controller stress test, not bulk damage.

The broader lesson is methodological: identifying an interpolating state map, suppressing its equilibrium tail, and validating its dynamics are independent problems. CALM solves the first two for the supplied two-surface model but does not cure the nonconservative force layering. A containment-converged study must therefore test stability anew, and a robust next-generation construction should evolve an electronic state or retain the switching derivatives rather than expecting controller shape alone to repair missing force consistency.

Reproducibility statement

The accompanying artifact contains the immutable potential files' checksums, the timestamped controller specification, complete configuration-level DFT evaluations, calibration and equilibrium trajectory ledgers, generated LAMMPS inputs, analysis programs, and figure sources. Numeric macros in this manuscript are generated from machine-readable result tables. Random seeds, software versions, missing-value rules, and the finite-volume deviation are stated in the methods and retained in the ledgers.

References

- Graeme J. Ackland. Temperature dependence in interatomic potentials and an improved potential for Ti. *Journal of Physics: Conference Series*, 402(1):012001, 2012. doi: 10.1088/1742-6596/402/1/012001.
- Ralf Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. *Physical Review B*, 99(1):014104, 2019. doi: 10.1103/PhysRevB.99.014104.
- Dorothy M. Duffy and A. M. Rutherford. Including the effects of electronic stopping and electron-ion interactions in radiation damage simulations. *Journal of Physics: Condensed Matter*, 19(1):016207, 2007. doi: 10.1088/0953-8984/19/1/016207.
- Tamar Flash and Neville Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *The Journal of Neuroscience*, 5(7):1688–1703, 1985. doi: 10.1523/JNEUROSCI.05-07-01688.1985.
- David Immel, Ralf Drautz, and Godehard Sutmann. Adaptive-precision potentials for large-scale atomistic simulations. *The Journal of Chemical Physics*, 162(11):114119, 2025. doi: 10.1063/5.0245877.
- S. Khakshouri, D. Alfè, and Dorothy M. Duffy. Development of an electron-temperature-dependent interatomic potential for molecular dynamics simulation of tungsten under electronic excitation. *Physical Review B*, 78(22):224304, 2008. doi: 10.1103/PhysRevB.78.224304.

- 689 Georg Kresse and Jürgen Furthmüller. Efficient iterative schemes for *Ab Initio* total-energy cal-
690 culations using a plane-wave basis set. *Physical Review B*, 54(16):11169–11186, 1996. doi:
691 10.1103/PhysRevB.54.11169.
- 692 Georg Kresse and Daniel Joubert. From ultrasoft pseudopotentials to the projector augmented-wave
693 method. *Physical Review B*, 59(3):1758–1775, 1999. doi: 10.1103/PhysRevB.59.1758.
- 694 Peter Mahler Larsen, Søren Schmidt, and Jakob Schiøtz. Robust structural identification via polyhe-
695 dral template matching. *Modelling and Simulation in Materials Science and Engineering*, 24(5):
696 055007, 2016. doi: 10.1088/0965-0393/24/5/055007.
- 697 Mihai-Cosmin Marinica, Lisa Ventelon, Mark R. Gilbert, Laurent Proville, Sergei L. Dudarev, Jaime
698 Marian, Guy Bencteux, and François Willaime. Interatomic potentials for modelling radiation
699 defects and dislocations in tungsten. *Journal of Physics: Condensed Matter*, 25(39):395502, 2013.
700 doi: 10.1088/0953-8984/25/39/395502.
- 701 N. David Mermin. Thermal properties of the inhomogeneous electron gas. *Physical Review*, 137
702 (5A):A1441–A1443, 1965. doi: 10.1103/PhysRev.137.A1441.
- 703 Samuel T. Murphy, Yvelin Giret, Szymon L. Daraszewicz, Anthony C. Lim, Alexander L. Shluger,
704 Katsumi Tanimura, and Dorothy M. Duffy. Contribution of electronic excitation to the structural
705 evolution of ultrafast laser-irradiated tungsten nanofilms. *Physical Review B*, 93(10):104105, 2016.
706 doi: 10.1103/PhysRevB.93.104105.
- 707 John P. Perdew, Kieron Burke, and Matthias Ernzerhof. Generalized gradient approximation made
708 simple. *Physical Review Letters*, 77(18):3865–3868, 1996. doi: 10.1103/PhysRevLett.77.3865.
- 709 Andrew Rohskopf, Caleb Sievers, Nicholas Lubbers, Mary Alice Cusentino, Jonathan Goff, Jan
710 Janssen, Michael McCarthy, David Montes Oca de Zapiain, Svetoslav Nikolov, Khachik Sargsyan,
711 Daniel Sema, Emily Sikorski, Laura Williams, Aidan P. Thompson, and Mitchell A. Wood.
712 FitSNAP: Atomistic machine learning with LAMMPS. *Journal of Open Source Software*, 8(84):
713 5118, 2023. doi: 10.21105/joss.05118.
- 714 Andrea E. Sand, Glen P. Kiely, Artur Tamm, and Alfredo A. Correa. Electronic effects in radiation-
715 induced collision cascades in nickel. *Physical Review Materials*, 9(5):053603, 2025. doi: 10.1103/
716 PhysRevMaterials.9.053603.
- 717 Alexander Stukowski. Visualization and analysis of atomistic simulation data with OVITO—the
718 open visualization tool. *Modelling and Simulation in Materials Science and Engineering*, 18(1):
719 015012, 2009. doi: 10.1088/0965-0393/18/1/015012.
- 720 Aidan P. Thompson, H. Metin Aktulga, Richard Berger, Dan S. Bolintineanu, W. Michael Brown,
721 Paul S. Crozier, Pieter J. in ’t Veld, Axel Kohlmeyer, Stan G. Moore, Trung Dac Nguyen, Ray
722 Shan, Mark J. Stevens, Julien Tranchida, Christian Trott, and Steven J. Plimpton. LAMMPS—a
723 flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum
724 scales. *Computer Physics Communications*, 271:108171, 2022. doi: 10.1016/j.cpc.2021.108171.
- 725 Mitchell A. Wood, Mary Alice Cusentino, Brian D. Wirth, and Aidan P. Thompson. Data-driven
726 material models for atomistic simulation. *Physical Review B*, 99(18):184305, 2019. doi: 10.1103/
727 PhysRevB.99.184305.
- 728 Mitchell A. Wood, Gabrielle A. Koknat, and Aidan P. Thompson. Beyond classical molecular
729 dynamics with layered interatomic potentials. Research Square preprint, 2026.
- 730 Eva Zarkadoula, Dorothy M. Duffy, Kai Nordlund, Michael A. Seaton, Ilian T. Todorov, William J.
731 Weber, and Kostya Trachenko. Electronic effects in high-energy radiation damage in tungsten.
732 *Journal of Physics: Condensed Matter*, 27(13):135401, 2015. doi: 10.1088/0953-8984/27/13/
733 135401.
- 734 Yanwen Zhang and William J. Weber. Ion irradiation and modification: The role of coupled electronic
735 and nuclear energy dissipation and subsequent nonequilibrium processes in materials. *Applied*
736 *Physics Reviews*, 7(4):041307, 2020. doi: 10.1063/5.0027462.
- 737 James F. Ziegler, M. D. Ziegler, and Jochen P. Biersack. SRIM—the stopping and range of ions in
738 matter (2010). *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions*
739 *with Materials and Atoms*, 268(11–12):1818–1823, 2010. doi: 10.1016/j.nimb.2010.02.091.